

Brian Yin

📞 647-673-9258 | ✉ brian.yin@mail.utoronto.ca | [in linkedin.com/in/brianyinn/](https://www.linkedin.com/in/brianyinn/) | github.com/yinbri

EDUCATION

University of Toronto

Bachelor of Computer Science

Toronto, ON

Anticipated Graduation: May 2028

– Cumulative GPA: 4.0/4.0

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript/TypeScript, SQL

Frameworks: Spring, Node.js, Angular, React.js, Flask, FastAPI, JUnit

Tools & Platforms: AWS, Kubernetes, Docker, Jenkins, Redis, DynamoDB, PostgreSQL, MongoDB

EXPERIENCE

Royal Bank of Canada

July 2025 – Aug 2025

Full-Stack Software Developer Intern

Toronto, ON

- Engineered a **Slack-integrated RAG chatbot (SRE Assist)** using **LangChain**, **GPT-4**, and **vector databases**, reducing application lookup time from **15 Minutes** → **5 seconds (99% faster)**
- Projected to save **2400+ engineer-hours annually** across a 10-person SRE team by streamlining incident resolution and knowledge retrieval
- Architected **backend features with PostgreSQL + Node.js**, and built **monitoring dashboards in React**, enabling real-time analytics of chatbot usage
- Outlined roadmap for **cross-department adoption**, customizing connectors to ServiceNow, IIPM, and other data sources to improve workflows across RBC

Royal Bank of Canada

July 2024 – Aug 2024

Full-Stack Software Developer Intern

Toronto, ON

- Designed and deployed an internal **bounty board platform** adopted by **50+ engineers**, increasing engagement in documentation and bug-fix tasks
- Implemented **backend GitHub REST API integration** for automated task creation, tracking, and repository linkage, reducing coordination overhead by **25%+**
- Accelerated backlog clearance, enabling “unwanted” tasks to be completed **up to 3× faster**, directly improving **client issue resolution speed**
- Collaborated in an **Agile team of 5**, delivering full-stack features and gaining hands-on experience with **REST API design**, **web development**, and iterative delivery

PROJECTS

Plant Trait Identification (CNN) | *Python, Keras, NumPy*



- Developed and trained a **CNN model** to classify plant traits with **92% accuracy**, supporting climate change research
- Preprocessed and augmented a dataset of **10,000+ plant images**, improving model robustness and reducing overfitting by **20%**
- Extracted insights on **leaf morphology**, **stem structure**, and **climate adaptation**, accelerating research workflows for environmental scientists

Cineplex MCP Server | *TypeScript, Node.js, MCP*



- Built an **MCP server** enabling Claude to query live showtimes and **score seat availability** against configurable quality constraints (row/side exclusion, contiguous-block requirements)
- Designed a **normalized data adapter layer** decoupling seat-scoring logic from upstream API response shape, plus an in-memory **caching/throttling layer** to keep request volume low

Student Engagement Rewards System | *Angular, Flask, MongoDB*



- Built a **full-stack points-based rewards platform** to track event participation and manage prize redemptions across **100+ students**
- Iterated with feedback from teachers and students, driving a **20%+ increase in event attendance**